

• Find The critical point & determine
The extreme Value :-
" Abs. Max. & Abs. min "

1) $f(x) = x^2 - 10$ on $[-2, 3]$

we must find critical point

$$f'(x) = 2x \Rightarrow 2x = 0 \Rightarrow \underline{\underline{x = 0}}$$

$\therefore x = 0$ is critical point

end point $-2, 3 \Rightarrow$

all point is $-2, 0, 3$

$$f(x) = -6, -10, -1$$

$\downarrow$
Abs. Min $\downarrow$
Abs. Max

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2) $f(x) = x^3 - 9x$, $[-7, 7]$

find critical point $f'(x) = x^2 - 9 = 0$

$\Rightarrow x = \sqrt{9} = \pm 3$. $\Rightarrow$ at $x = -3, x = 3$ have a critical pt

$\Rightarrow$ end point of I -interval $\Rightarrow x = -7, x = 7$

all point are: $-7, -3, 3, 7$

$f(x)$: $-51\frac{1}{3}, 18, -18, 51\frac{1}{3}$
 $\downarrow$ Abs. Min. $\downarrow$ Abs. Max.

3) $f(x) = 1 - x^{\frac{1}{3}}$, $[-1, 8]$

critical point $f'(x) = -\frac{2}{3}x^{-\frac{1}{3}} = \frac{-2}{3\sqrt[3]{x}}$

$f'(x) \neq 0$

$f'(x) = \text{D.N.E} = \frac{\text{Value}}{0} \Rightarrow 3\sqrt[3]{x} = 0 \Rightarrow x = 0$ critical point

$\Rightarrow$ end point on $I \Rightarrow x = -1, x = 8$

all point are: $-1, 0, 8$

$f(x)$: $0, 1, -3$
 $\downarrow$ Abs. Max. $\downarrow$ Abs. Min.

4) $f(x) = x^2\sqrt{x+1}$, $[-1, 1]$

$f'(x) = 2x\sqrt{x+1} + \frac{x^2}{2\sqrt{x+1}} = \frac{1}{2\sqrt{x+1}}(4x\sqrt{x+1}\cdot\sqrt{x+1} + x^2)$

$= \frac{4x(x+1) + x^2}{2\sqrt{x+1}} = \frac{5x^2 + 4x}{2\sqrt{x+1}}$

$f'(x) = 0 \Rightarrow \frac{5x^2 + 4x}{2\sqrt{x+1}} = 0 \Rightarrow 5x^2 + 4x = 0 \Rightarrow x = 0, x = -\frac{4}{5}$

end point on $I \Rightarrow x = -1, x = 1$

all point are: $-1, -\frac{4}{5}, 0, 1$

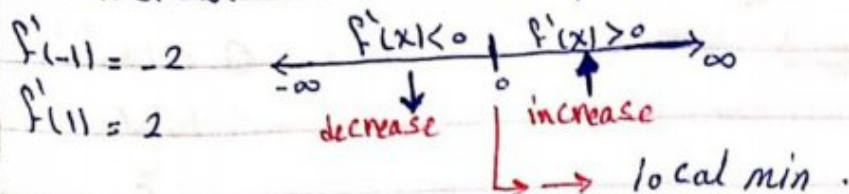
$f(x)$: $0, 0.286, 0, \sqrt{2} \approx 1.41$
 $\downarrow$ Abs. Min. $\downarrow$ Abs. Max.

Find The interval of increase & decrease :-

① $f(x) = x^2 - 16$

• find critical point $\Rightarrow f'(x) = 2x \Rightarrow 2x = 0 \rightarrow "x = 0"$

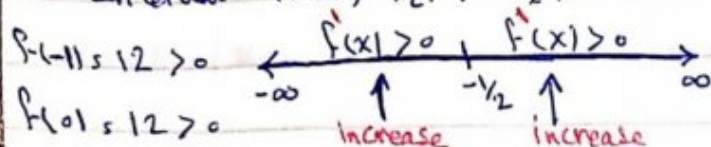
Interval $(-\infty, 0)$ $(0, \infty)$



② $f(x) = 12 - x - x^2$

• find critical point $\Rightarrow f'(x) = -1 - 2x = 0 \Rightarrow "x = -\frac{1}{2}"$

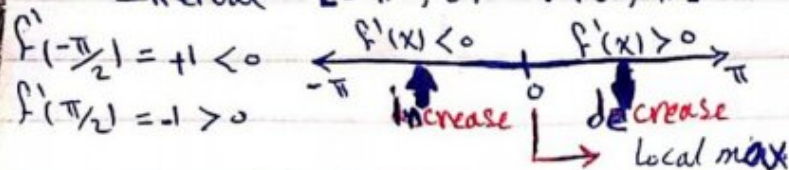
Interval $(-\infty, -\frac{1}{2})$ $(-\frac{1}{2}, \infty)$



③ $f(x) = \cos x$; $[-\pi, \pi]$

• find critical point $\Rightarrow f'(x) = -\sin x = 0 \Rightarrow "x = 0"$

Interval $[-\pi, 0)$ $(0, \pi]$

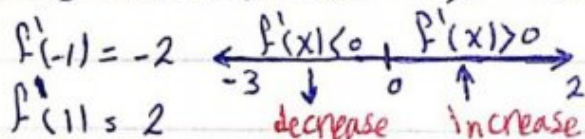


Find the Interval of increase & decrease , ② local extreme point

③ Abs. Max & Abs. Min by use first derivative

① $f(x) = x^2 + 3$; $[-3, 2]$

① $f'(x) = 2x \Rightarrow "x = 0"$ critical point



② local extreme point at $x = 0$ is a local min point.

③ all point -3 , 0 , 2

$f(x)$ 12 , 3 , 7
↓
Abs. Max ↓ Abs. Min

$$f(x) = x^{2/3}(x-5) \quad ; \quad [-5, 5]$$

$$f'(x) = \frac{2}{3}x^{-1/3}(x-5) + x^{2/3} = \frac{5}{3}x^{2/3} - \frac{10}{3}x^{-1/3}$$

$$= x^{1/3} \left(\frac{5x-10}{3} \right) = \frac{5x-10}{3x^{1/3}}$$

$$f'(x) = 0 \Rightarrow 5x - 10 = 0 \Rightarrow \underline{x = 2}$$

$$f'(x) = D.N.E \Rightarrow 3\sqrt[3]{x} = 0 \Rightarrow \underline{x = 0}$$

Critical point at $x = 0, 2$.

$$f'(-1) = 5 > 0 \quad \leftarrow f' > 0 \quad f' < 0 \quad f' > 0 \quad \rightarrow$$

$$f'(1) = -\frac{5}{3} < 0$$

$$f'(3) = \text{+ve} > 0$$

• at $x = 0$

has local max

increase decrease increase

at $x = 2$

has local min.

To find Abs. max & Abs. min

all point are : $-5, 0, 2, 5$

$$f(x) : -29.24, 0, -4.76, 0$$

Abs. Min

Abs. Max.

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Discuss the concavity & inflection :-

① $f(x) = x^4 - 2x^3 + 1$ s.h

$f'(x) = 4x^3 - 6x^2$

$f''(x) = 12x^2 - 12x = 0 \Rightarrow x(2x - 1) = 0 \Rightarrow x = 0, x = 1$

Interval $(-\infty, 0)$ $(0, 1)$ $(1, \infty)$

Sign of f'' $+ve$ $-ve$ $+ve$

Concave up $down$ up

at $x = 0, 1$ have inflection point

inflection point $\Rightarrow f(0) = 1 \Rightarrow (0, 1)$ & $(1, 0)$

$f(1) = 0$

② $f(x) = 2x^3 - 3x^2 + 12$

$f'(x) = 6x^2 - 6x$

$f''(x) = 12x - 6 \Rightarrow 12x - 6 = 0 \Rightarrow x = 1/2$

Interval $(-\infty, 1/2)$ $(1/2, \infty)$

Sign f'' $-ve$ $+ve$

Concave $down$ up

inflection point $\Rightarrow$ at $x = 1/2 \Rightarrow f(1/2) = 1/4 - 3/4 + 12 = 11 1/2$

$f(1/2) = 11 1/2 \Rightarrow$ inflection point is $(1/2, 11.5)$

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